Lord of the Wings [2026]

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Part I

Story section

I told Frodoff, "Come on. We need to leave. Now."

Questions could wait until we were on the road. Fortunately, he trusted me enough not to ask why. That was probably for the best. Explaining that he had accidentally accepted a rather unfortunate destiny by picking up a suspicious-looking stone was not the sort of conversation one had before breakfast.

Before we left Quackshire, there was one last thing we had to check: the village defences.

Halfbits were delightful folk: brave when necessary, generous by nature, and spectacularly relaxed about routine maintenance. That was rarely a good combination when the village defences depended on ancient magic.

Many years earlier, we had planted a forest of elven pines around the village. Their pine cones could be arranged into enchanted formations that repelled hostile magical creatures. The outposts around the village were arranged in three defensive rings, each using a different type of formation. In theory, every outpost should have been ready. In practice, this was Quackshire.

Each outpost had to build exactly one formation using all of its enchanted pine cones. The first type of protective formation was simple. The pine cones had to completely fill a square on a grid. The smallest valid formation was therefore a 1 x 1 square, requiring just `1` pine cone. Larger squares worked in exactly the same way:

```
1 x 1     2 x 2      3 x 3  ...

#         ##         ###
          ##         ###
                     ###
```

We inspected every outpost in the first defensive ring. Unsurprisingly, quite a few pine cones had long since lost their magic and needed replacing. We wrote down how many enchanted pine cones each outpost still had (your notes), then set out to **collect exactly enough fresh ones for every outpost to build a complete square using all of its pine cones**.

For example, if an outpost had `10` enchanted pine cones, that was one too many for a 3 x 3 square, but six too few for a 4 x 4 square. The outpost therefore had to collect exactly `6` new pine cones: `10 + 6 = 16`.

Example based on the following notes:

`3,11,12,16,24,25`

- The first outpost had `3` enchanted pine cones, so it had to collect `1` more to build a 2 x 2 square.
- The second outpost had `11` pine cones and needed `5` more to complete a 4 x 4 square.
- The third outpost had `12` pine cones and needed `4` more to complete a 4 x 4 square.
- The fourth outpost already had exactly `16` pine cones, enough to build a 4 x 4 square.
- The fifth outpost had `24` pine cones and needed `1` more to complete a 5 x 5 square.
- The final outpost already had exactly `25` pine cones, enough to build a 5 x 5 square.

In total, the outposts had to collect 1 + 5 + 4 + 0 + 1 + 0 = `11` new pine cones.

How many new pine cones must be collected so that every outpost can build a complete square using all of its pine cones?

Your notes for this part:

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Part I solved with answer: 1375

Check your progress

Part II

Once the first defensive ring had been secured, we continued to inspect the second one. Each outpost in this ring had to **build a square frame instead of a filled square.**

As a result, each formation required fewer enchanted pine cones. **The frame also had to surround at least one empty cell**; otherwise, the magic would not work. The smallest possible formation was therefore a 3 x 3 frame, requiring `8` pine cones. Larger square frames worked in the same way:

```
3 x 3     4 x 4      5 x 5  ...

###       ####       #####
#.#       #..#       #...#
###       #..#       #...#
          ####       #...#
                     #####
```

Once again, we noted how many enchanted pine cones each outpost still had (your notes). We then calculated how many additional pine cones had to be collected for every outpost to build a valid square frame using all of its pine cones.

Example based on the following notes:

`3,11,12,16,24,25`

- The first outpost had `3` enchanted pine cones, so it had to collect `5` more to build a 3 x 3 frame.
- The second outpost had `11` pine cones and needed `1` more to complete a 4 x 4 frame.
- The next three outposts already had exactly `12` pine cones for a 4 x 4 frame, `16` for a 5 x 5 frame, and `24` for a 7 x 7 frame, so they did not need to collect any more.
- The final outpost had `25` pine cones. The next valid formation was an 8 x 8 frame requiring 28 pine cones, so it had to collect `3` more.

In total, the outposts had to collect 5 + 1 + 0 + 0 + 0 + 3 = `9` new pine cones.

How many new pine cones must be collected so that every outpost can build a valid square frame using all of its pine cones?

Your notes for this part:

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Part II solved with answer: 332

Check your progress

Part III

Finally, we reached the outposts in the third defensive ring. Their formation looked much like the original **filled square, but with exactly one empty cell in its centre.**

This meant that only squares with an odd side length could be used. A square with an even side length had no single central cell, and the magic refused to accept an approximate centre.

The smallest valid formation was again 3 x 3 and required `8` pine cones. Larger formations followed the same rule:

```
3 x 3     5 x 5      7 x 7    ...

###       #####      #######
#.#       #####      #######
###       ##.##      #######
          #####      ###.###
          #####      #######
                     #######
                     #######
```

We noted how many enchanted pine cones remained at each outpost (your notes). As before, we had to collect exactly enough new ones for every outpost to build a valid formation using all of its pine cones.

Example based on the following notes:

`3,11,12,16,24,25`

- The first outpost had `3` enchanted pine cones, so it had to collect `5` more for a 3 x 3 formation.
- The second outpost had `11` pine cones. The next valid formation was 5 x 5 and required 24 pine cones, so it had to collect `13` more.
- The third outpost had `12` pine cones and needed another `12` to complete a 5 x 5 formation.
- The fourth outpost had `16` pine cones and needed `8` more for a 5 x 5 formation.
- The fifth outpost already had exactly `24` pine cones, enough for a 5 x 5 formation.
- The final outpost had `25` pine cones. The next valid formation was 7 x 7 and required 48 pine cones, so it had to collect `23` more.

In total, the outposts had to collect 5 + 13 + 12 + 8 + 0 + 23 = `61` new pine cones.

How many new pine cones must be collected so that every outpost can build a complete square with exactly one empty cell in its centre, using all of its pine cones?

Your notes for this part:

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Part III solved with answer: 641970

Check your progress

Puzzle solved! Don't stop now!

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